



QSS: Analysis of HBB99 protocol

Quantum Computing

MSc in Cybersecurity Engineering

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February 19th, 2026



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di Torino**



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Secret Sharing

1 Introduction

- **Secret sharing** (SS) is a cryptographic method used to breakdown a secret into multiple parts (**shares**)
- Secret can be reconstructed **if and only if** enough parties join the process
- The problem:

“Alice wants to have an action taken on her behalf, but she does not trust the single agents Bob and Charlie, therefore, they must carry the action together”



Quantum Secret Sharing

1 Introduction

- Main problem of SS is **eavesdropping**
- Using **quantum cryptographic protocols** we can fix the problem
- Two approaches:
 - **Embed classical SS** procedures in quantum cryptographic protocols
 - Use **quantum mechanics to do both** the information splitting and the eavesdropping protection

The latter is more interesting and achievable with **multiparticle entanglement**



Overview

1 Introduction

- The presentation will cover the implementation of HBB99 on the Lagrange, analyzing also its security
 1. **The HBB99 Protocol**
Brief analysis of the protocol and its implementation on Lagrange machine
 2. **Dishonest Participant Analysis**
Analysis of the conditions when an insider attack becomes possible, and practical demonstration
 3. **Fairness Fix**
Patching the protocol adding a commit-then-reveal approach



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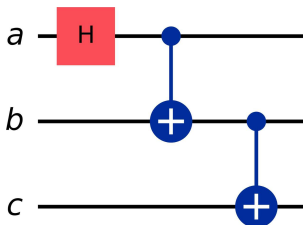
► Conclusions



GHZ State

2 HBB99: Protocol Analysis

- **GHZ state** as starting point for the protocol



- Mathematically, a GHZ state of n particles is

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle^{\otimes n} + |1\rangle^{\otimes n} \right)$$



Measurement Correlations (1/3)

2 HBB99: Protocol Analysis

- Every party measures his qubit choosing randomly between two possible directions, or **measurement bases (MBs)**

$$M_x = M_z H \qquad M_y = M_z H S^\dagger$$

- Therefore, the particle can be in the following **eigenstates**

$$\begin{aligned} |x^+\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) & |y^+\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) \\ |x^-\rangle &= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) & |y^-\rangle &= \frac{1}{\sqrt{2}} (|0\rangle - i|1\rangle) \end{aligned}$$



Measurement Correlations (2/3)

2 HBB99: Protocol Analysis

- Noting that

$$|0\rangle = \frac{1}{\sqrt{2}}(|x^+\rangle + |x^-\rangle) = \frac{1}{\sqrt{2}}(|y^+\rangle + |y^-\rangle)$$

$$|1\rangle = \frac{1}{\sqrt{2}}(|x^+\rangle - |x^-\rangle) = \frac{-i}{\sqrt{2}}(|y^+\rangle - |y^-\rangle)$$

- Given a GHZ state, we can **compute the expected state for the last qubit**
- If Alice (a) and Bob (b) both do M_x , Charlie (c) will have

$$\begin{aligned} |\psi\rangle = \frac{1}{\sqrt{8}} & \left[(|x^+\rangle_a |x^+\rangle_b + |x^-\rangle_a |x^-\rangle_b) (|0\rangle_c + |1\rangle_c) \right. \\ & \left. + (|x^+\rangle_a |x^-\rangle_b + |x^-\rangle_a |x^+\rangle_b) (|0\rangle_c - |1\rangle_c) \right] \end{aligned}$$



Measurement Correlations (3/3)

2 HBB99: Protocol Analysis

- Charlie can determine if the results of Alice and Bob differ by performing a **measurement with the right MB**
- The table summarizes the effects of Alice's and Bob's measurements on Charlie's qubit

	$ x^+\rangle_b$	$ x^-\rangle_b$	$ y^+\rangle_b$	$ y^-\rangle_b$
$ x^+\rangle_a$	$ x^+\rangle_c$	$ x^-\rangle_c$	$ y^-\rangle_c$	$ y^+\rangle_c$
$ x^-\rangle_a$	$ x^-\rangle_c$	$ x^+\rangle_c$	$ y^+\rangle_c$	$ y^-\rangle_c$
$ y^+\rangle_a$	$ y^-\rangle_c$	$ y^+\rangle_c$	$ x^-\rangle_c$	$ x^+\rangle_c$
$ y^-\rangle_a$	$ y^+\rangle_c$	$ y^-\rangle_c$	$ x^+\rangle_c$	$ x^-\rangle_c$



Logical interpretation (1/2)

2 HBB99: Protocol Analysis

- The **number of M_y performed** ($|M_y|$) determines the success of the procedure
- Given a GHZ state composed by n qubits, $|M_y|$ **must be even**, obtaining as **final behaviour**

$$\begin{cases} q_n = q_1 \oplus q_2 \oplus \dots \oplus q_{n-1} & \text{if } |M_y| \equiv 0 \pmod{4} \\ q_n = \neg(q_1 \oplus q_2 \oplus \dots \oplus q_{n-1}) & \text{if } |M_y| \equiv 2 \pmod{4} \end{cases}$$

- If $|M_y|$ is not even, the behaviour of the particle will be the one of $H^{\otimes n} |0\rangle^{\otimes n}$



Logical interpretation (2/2)

2 HBB99: Protocol Analysis

- The reason is the **induced phase of $-i$** when M_y is performed
- If $|M_y| \equiv 0 \pmod{4}$
 - Global **phase change of $(-i)^{4k} = 1$**
 - Last measurement will make the GHZ state collapse to an even parity bit string
- If $|M_y| \equiv 2 \pmod{4}$
 - Global **phase change of $(-i)^{4k+2} = -1$**
 - Last measurement will make the GHZ state collapse to an odd parity bit string



Protocol Flow: Initialization

2 HBB99: Protocol Analysis

1. Initialization:

- The **dealer** creates m GHZ states of n particles
- For each GHZ state, the dealer keeps the first qubit q_1 , and distributes to **each** i^{th} **party** the i^{th} qubit q_i

2. Measurement Phase:

- Each party measures their qubit, **randomly choosing between the X or Y basis**
- They record their choice and measurement outcome (0 or 1)

3. Sifting (Basis Announcement):

- Parties broadcast their basis choices
- They keep only the N rounds where $|M_y|$ is even



Protocol Flow: Eavesdropping Check

2 HBB99: Protocol Analysis

4. Eavesdropping Check:

- Dealer selects a random subset of valid rounds to be **j test rounds**
- All parties broadcast their measurement outcomes for these rounds, verifying if the parity constraint holds:

$$\begin{cases} q_1 \oplus q_2 \oplus \dots \oplus q_n = 0 & \text{if } |M_Y| \equiv 0 \pmod{4} \\ q_1 \oplus q_2 \oplus \dots \oplus q_n = 1 & \text{if } |M_Y| \equiv 2 \pmod{4} \end{cases}$$

- If the error rate exceeds a **threshold**, the protocol aborts



Protocol Flow: Key Generation

2 HBB99: Protocol Analysis

5. Data Phase (Split Key Creation):

- Unchecked rounds are used to create the split key
- Dealer's results become the **secret key**

$$D = \{q_{1,1}, \dots, q_{1,N-j}\}$$

- Parties hold the **shares** S_i such that:

$$D = \bigoplus_{i=2}^n S_i \oplus (\text{Parity Correction})$$

so they must cooperate to recover D



Demo 1: HBB99 with 3 participants

2 HBB99: Protocol Analysis

- **Roles:** Alice is the Dealer; Bob and Charlie are the shareholders
- **Configuration:**
 - **Target:** Create a split key of 8 bits
 - **Security:** 50% of the valid rounds are used as test rounds, and the threshold is set to 25%
- **NISQ Adaptation:**
 - Unlike optical implementations, the IQM Spark is a noisy superconducting processor.
 - We select the measurement of the run with highest count to mitigate readout noise.
 - This reduces the baseline error to near-zero



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The Scenario

3 HBB99: Dishonest Participant

- **Scenario:**

- Charlie* is the dishonest insider
- He possesses his own legitimate particle (c)
- Crucially, he **intercepts the particle directed to Bob** (b), gaining physical control over 2 of the 3 GHZ qubits

- **Charlie*'s Objectives:**

- **Evade Detection:** Manipulate the state such that the error rate in the test rounds remains below the threshold, fooling Alice and Bob into believing the channel is secure
- **Reconstruct the Secret:** Perform a sequence of action that let him retrieve all the secret bits of the other parties, without any cooperation



Intercept-Resend Attack (1/2)

3 HBB99: Dishonest Participant

- Charlie* does not know which basis Alice used (X or Y), so he is forced to **guess Alice's measurement basis (MB)**

$$\text{Basis X : } \frac{|00\rangle \pm |11\rangle}{\sqrt{2}} \quad \text{or} \quad \text{Basis Y : } \frac{|00\rangle \pm i|11\rangle}{\sqrt{2}}$$

- This measurement immediately **collapses the quantum state**, destroying the original entanglement



Intercept-Resend Attack (2/2)

3 HBB99: Dishonest Participant

- Bob receives a **resend particle** prepared by Charlie* based on his measurement result
 - **The State:** Depending on Charlie*'s guess, Bob receives a deterministic eigenstate $|x^\pm\rangle$ or $|y^\pm\rangle$.
 - **The Mismatch:** If Charlie* guessed wrong (e.g., sent X, Bob checks Y), Bob measures a superposition

$$|x^\pm\rangle = \frac{1}{\sqrt{4}}((1 \mp i)|y^+\rangle + (1 \pm i)|y^-\rangle) \quad |y^\pm\rangle = \frac{1}{\sqrt{4}}((1 \pm i)|x^+\rangle + (1 \mp i)|x^-\rangle)$$

- **Detection Probability:** The attack fails if Charlie* guesses the wrong basis AND the result is wrong

$$P(\text{Error}) = P(\text{Wrong MB}) \times P(\text{Wrong Result}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$



Demo 2: Intercept-Resend Attack

3 HBB99: Dishonest Participant

- **Roles:** Alice is the Dealer; Bob is the victim; Charlie* is the active attacker
- **Configuration:**
 - **Attack Vector:** Charlie* measures the GHZ particles immediately in a random basis (X or Y) and resends a fresh particle to Bob.
 - **Target:** Demonstrate the detectability of this naive attack and the incapability of retrieving the secret
- **Results:**
 - The error rate is typically higher, and Charlie* cannot recover Alice's secret
 - **Outcome:** The protocol has high probability of being aborted, and if that is not the case, the deduced key of Alice is likely wrong



Fake Entanglement Attack (1/2)

3 HBB99: Dishonest Participant

- The attack exploits the **order of bases announcement**:
- Charlie* uses **two other particles** b' and c' to create a Bell state

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{b'c'} - |11\rangle_{b'c'})$$

sending b' to Bob

- Everyone measures their particle, but **Charlie* waits**



Fake Entanglement Attack (2/2)

3 HBB99: Dishonest Participant

- Bob's result is uncorrelated with a , but has a particular **relation with c'**

$$M_x^{b'} |\Phi^-\rangle = \frac{1}{\sqrt{4}} \left[|x^+\rangle (|0\rangle_{c'} - |1\rangle_{c'}) + |x^-\rangle (|0\rangle_{c'} + |1\rangle_{c'}) \right]$$

$$M_y^{b'} |\Phi^-\rangle = \frac{1}{\sqrt{4}} \left[|y^+\rangle (|0\rangle_{c'} + i|1\rangle_{c'}) + |y^-\rangle (|0\rangle_{c'} - i|1\rangle_{c'}) \right]$$

- Charlie* can **retrieve the secrets of Alice and Bob**
 - Alice: Measurements of b and c , related to the one of a
 - Bob: Measurement of c' , related to the one of b'
- Charlie* announces a **compatible direction** and **computes the secret** to obtain the expected parity



Demo 3: Fake Entanglement Attack

3 HBB99: Dishonest Participant

- **Roles:** Charlie* acts as a sophisticated attacker creating an additional entangled pair
- **Configuration:**
 - **Attack Vector:** Charlie* uses 2 additional qubits to create a fake Bell pair for Bob
 - **Target:** Reconstruct the secret key without triggering the error threshold
- **Results:**
 - The error rate in Test Rounds remains consistent with the baseline
 - **Outcome:** The protocol continues and Charlie* successfully computes the key of Alice, fully compromising the security



Ancilla Attack: Avoid Detection (1/4)

3 HBB99: Dishonest Participant

- Charlie* makes an **ancilla** e interact with the received particles, so the state becomes

$$|\psi\rangle_{abce} = \sum_{i,j=0}^1 a_{ij} |ij\rangle_{ab} |\varepsilon_{ij}\rangle_{ce}$$

- Charlie* **waits to discover Alice's and Bob's MBs**, declaring a MB that will not be immediately used
- Let's consider the case where both Alice and Bob measure along x direction



Ancilla Attack: Avoid Detection (2/4)

3 HBB99: Dishonest Participant

- The state becomes

$$\begin{aligned} |\psi\rangle_{abce} = \frac{1}{\sqrt{4}} & \left[|x^+\rangle_a |x^+\rangle_b (a_{00} |\varepsilon_{00}\rangle + a_{01} |\varepsilon_{01}\rangle + a_{10} |\varepsilon_{10}\rangle + a_{11} |\varepsilon_{11}\rangle)_{ce} \right. \\ & + |x^+\rangle_a |x^-\rangle_b (a_{00} |\varepsilon_{00}\rangle - a_{01} |\varepsilon_{01}\rangle + a_{10} |\varepsilon_{10}\rangle - a_{11} |\varepsilon_{11}\rangle)_{ce} \\ & + |x^-\rangle_a |x^+\rangle_b (a_{00} |\varepsilon_{00}\rangle + a_{01} |\varepsilon_{01}\rangle - a_{10} |\varepsilon_{10}\rangle - a_{11} |\varepsilon_{11}\rangle)_{ce} \\ & \left. + |x^-\rangle_a |x^-\rangle_b (a_{00} |\varepsilon_{00}\rangle - a_{01} |\varepsilon_{01}\rangle - a_{10} |\varepsilon_{10}\rangle + a_{11} |\varepsilon_{11}\rangle)_{ce} \right] \end{aligned}$$

- Writing it in a more compact way

$$\begin{aligned} |\psi\rangle_{abce} = \frac{1}{\sqrt{4}} & \left[|x^+\rangle_a |x^+\rangle_b |\varphi_{x^+x^+}\rangle + |x^+\rangle_a |x^-\rangle_b |\varphi_{x^+x^-}\rangle \right. \\ & \left. + |x^-\rangle_a |x^+\rangle_b |\varphi_{x^-x^+}\rangle + |x^-\rangle_a |x^-\rangle_b |\varphi_{x^-x^-}\rangle \right] \end{aligned}$$



Ancilla Attack: Avoid Detection (3/4)

3 HBB99: Dishonest Participant

- To avoid being found out, Charlie* must distinguish two sets:
 - $\{|\varphi_{x+x+}\rangle, |\varphi_{x-x-}\rangle\}$, i.e., the states where Charlie* should measure $|x^+\rangle$
 - $\{|\varphi_{x+x-}\rangle, |\varphi_{x-x+}\rangle\}$, i.e., the states where Charlie* should measure $|x^-\rangle$
- Two states can be distinguished clearly if they are orthogonal, i.e.

$$\begin{cases} \langle \varphi_{x+x+} | \varphi_{x+x-} \rangle = 0 \\ \langle \varphi_{x+x+} | \varphi_{x-x+} \rangle = 0 \\ \langle \varphi_{x-x-} | \varphi_{x+x-} \rangle = 0 \\ \langle \varphi_{x-x-} | \varphi_{x-x+} \rangle = 0 \end{cases}$$



Ancilla Attack: Avoid Detection (4/4)

3 HBB99: Dishonest Participant

- By computing the system for every combination of measurements over a and b , the **final constraints** are

$$\begin{cases} a_{00}^* a_{01} \langle \varepsilon_{00} | \varepsilon_{01} \rangle = a_{00}^* a_{10} \langle \varepsilon_{00} | \varepsilon_{10} \rangle = 0 \\ a_{00}^* a_{11} \langle \varepsilon_{00} | \varepsilon_{11} \rangle = a_{01}^* a_{10} \langle \varepsilon_{01} | \varepsilon_{10} \rangle = 0 \\ a_{01}^* a_{11} \langle \varepsilon_{01} | \varepsilon_{11} \rangle = a_{10}^* a_{11} \langle \varepsilon_{10} | \varepsilon_{11} \rangle = 0 \\ |a_{00}| = |a_{11}| \\ |a_{01}| = |a_{10}| \end{cases}$$

- This means that Charlie* can escape detection by building a state that satisfies those equations



Ancilla Attack: Retrieving Alice's Secret (1/3)

3 HBB99: Dishonest Participant

- Charlie* now must measure his qubits to deduce Alice's secret
- Two cases:
 - **Alice has x^+** : Charlie* has a mix of the states $|\varphi_{x^+x^+}\rangle$ and $|\varphi_{x^+x^-}\rangle$

$$\rho_{x^+} = \frac{1}{2}(|\varphi_{x^+x^+}\rangle \langle \varphi_{x^+x^+}| + |\varphi_{x^+x^-}\rangle \langle \varphi_{x^+x^-}|)$$

- **Alice has x^-** : Charlie* has a mix of the states $|\varphi_{x^-x^+}\rangle$ and $|\varphi_{x^-x^-}\rangle$

$$\rho_{x^-} = \frac{1}{2}(|\varphi_{x^-x^+}\rangle \langle \varphi_{x^-x^+}| + |\varphi_{x^-x^-}\rangle \langle \varphi_{x^-x^-}|)$$

- Charlie* should be able to **distinguish ρ_{x^+} and ρ_{x^-}**



Ancilla Attack: Retrieving Alice's Secret (2/3)

3 HBB99: Dishonest Participant

- Generally, to discriminate two mixed states ρ_1 and ρ_2 occurring with probability p_1 and p_2 , the **minimum error probability attainable** is

$$P_E = \frac{1}{2} - \frac{1}{2} ||p_2 \rho_2 - p_1 \rho_1||$$

- The final probability obtained for all four cases is

$$P_E = \frac{1}{2}(1 - 4|a_{00}||a_{10}|)$$



Ancilla Attack: Retrieving Alice's Secret (3/3)

3 HBB99: Dishonest Participant

- The mutual information between Alice and Charlie* in terms of **binary entropy function** $H(p)$ is

$$I_{AC} = 1 - H(P_E)$$

- The **only way to maximize** it is to have $|a_{00}| |a_{10}|$ equal to $\frac{1}{4}$, obtaining a P_E of 0, i.e., we need **both the norms equal to** $\frac{1}{2}$
- Therefore the **final requirements** will be

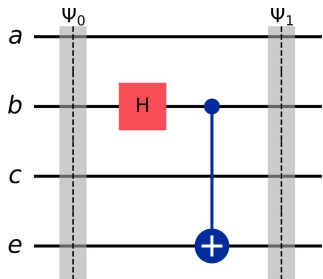
$$\begin{cases} \langle \varepsilon_{00} | \varepsilon_{01} \rangle = \langle \varepsilon_{00} | \varepsilon_{10} \rangle = \langle \varepsilon_{00} | \varepsilon_{11} \rangle = 0 \\ \langle \varepsilon_{01} | \varepsilon_{10} \rangle = \langle \varepsilon_{01} | \varepsilon_{11} \rangle = \langle \varepsilon_{10} | \varepsilon_{11} \rangle = 0 \\ |a_{00}| = |a_{01}| = |a_{10}| = |a_{11}| = \frac{1}{2} \end{cases}$$



Ancilla Attack: Practical Implementation (1/8)

3 HBB99: Dishonest Participant

- Just an ancilla e in the state $|0\rangle$ is needed
- Charlie* performs the following interaction between b and e



$$|\Psi_0\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle)_{abc} \otimes |0\rangle_e$$

$$|\Psi_1\rangle = \frac{1}{2} (|00\rangle_{ab} |00\rangle_{ce} + |01\rangle_{ab} |01\rangle_{ce} + |10\rangle_{ab} |10\rangle_{ce} - |11\rangle_{ab} |11\rangle_{ce})$$

- Let's see how Charlie*'s state is if he waits, while the other measure



Ancilla Attack: Practical Implementation (2/8)

3 HBB99: Dishonest Participant

(i) Alice and Bob do both M_x

$$\begin{aligned}|\varphi_{x^+x^+}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)_{ce} \\|\varphi_{x^+x^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)_{ce} \\|\varphi_{x^-x^+}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle)_{ce} \\|\varphi_{x^-x^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)_{ce}\end{aligned}$$

(ii) Alice does M_x and Bob does M_y

$$\begin{aligned}|\varphi_{x^+y^+}\rangle &= \frac{1}{2}(|00\rangle - i|01\rangle + |10\rangle + i|11\rangle)_{ce} \\|\varphi_{x^+y^-}\rangle &= \frac{1}{2}(|00\rangle + i|01\rangle + |10\rangle - i|11\rangle)_{ce} \\|\varphi_{x^-y^+}\rangle &= \frac{1}{2}(|00\rangle - i|01\rangle - |10\rangle - i|11\rangle)_{ce} \\|\varphi_{x^-y^-}\rangle &= \frac{1}{2}(|00\rangle + i|01\rangle - |10\rangle + i|11\rangle)_{ce}\end{aligned}$$



Ancilla Attack: Practical Implementation (3/8)

3 HBB99: Dishonest Participant

(iii) Alice does M_y and Bob does M_x

$$\begin{aligned}|\varphi_{y^+x^+}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle - i|10\rangle + i|11\rangle)_{ce} \\|\varphi_{y^+x^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle - i|10\rangle - i|11\rangle)_{ce} \\|\varphi_{y^-x^+}\rangle &= \frac{1}{2}(|00\rangle + i|01\rangle + i|10\rangle - |11\rangle)_{ce} \\|\varphi_{y^-x^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle + i|10\rangle + i|11\rangle)_{ce}\end{aligned}$$

(iv) Alice and Bob do both M_y

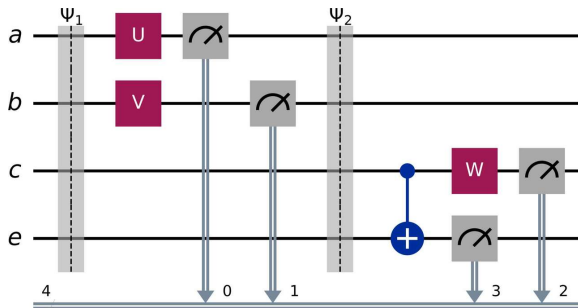
$$\begin{aligned}|\varphi_{y^+y^+}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle - i|10\rangle + i|11\rangle)_{ce} \\|\varphi_{y^+y^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle - i|10\rangle - i|11\rangle)_{ce} \\|\varphi_{y^-y^+}\rangle &= \frac{1}{2}(|00\rangle + i|01\rangle + i|10\rangle - |11\rangle)_{ce} \\|\varphi_{y^-y^-}\rangle &= \frac{1}{2}(|00\rangle - |01\rangle + i|10\rangle + i|11\rangle)_{ce}\end{aligned}$$



Ancilla Attack: Practical Implementation (4/8)

3 HBB99: Dishonest Participant

- Charlie*'s behaviour during **test rounds**



	(i)	(ii)	(iii)	(iv)
U	H	H	$HS^\dagger$	$HS^\dagger$
V	H	$HS^\dagger$	H	$HS^\dagger$
W	H	$HS^\dagger$	$HS^\dagger$	H

- Let's analyse what happens in case (i)



Ancilla Attack: Practical Implementation (5/8)

3 HBB99: Dishonest Participant

- Charlie* needs to discriminate $\{|\varphi_{x^+x^+}\rangle, |\varphi_{x^-x^-}\rangle\}$ and $\{|\varphi_{x^+x^-}\rangle, |\varphi_{x^-x^+}\rangle\}$

$$H_c(CX|\varphi_{x^+x^+}\rangle) = H_c(CX(\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)_{ce})) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)_{ce}$$

$$H_c(CX(|\varphi_{x^+x^-}\rangle) = H_c(CX(\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle + |11\rangle)_{ce})) = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)_{ce}$$

$$H_c(CX(|\varphi_{x^-x^+}\rangle) = H_c(CX(\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle)_{ce})) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{ce}$$

$$H_c(CX(|\varphi_{x^-x^-}\rangle) = H_c(CX(\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)_{ce})) = \frac{1}{\sqrt{2}}(-|01\rangle + |10\rangle)_{ce}$$

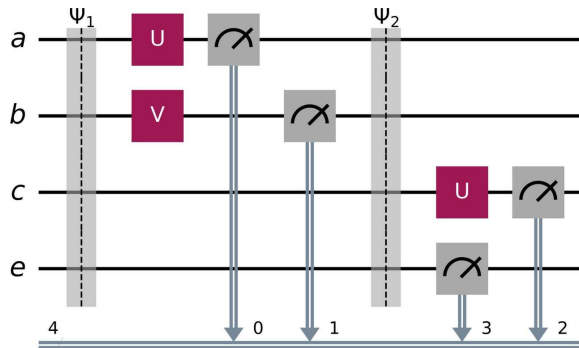
- Charlie* will say $|x^+\rangle$ when $|00\rangle$ or $|11\rangle$ is measured, while he will say $|x^-\rangle$ when $|01\rangle$ or $|10\rangle$ is measured



Ancilla Attack: Practical Implementation (6/8)

3 HBB99: Dishonest Participant

- Charlie*'s behaviour during **data rounds**



	(i)	(ii)	(iii)	(iv)
U	H	H	$HS^\dagger$	$HS^\dagger$
V	H	$HS^\dagger$	H	$HS^\dagger$

- Let's analyse what happens in case (i)



Ancilla Attack: Practical Implementation (7/8)

3 HBB99: Dishonest Participant

- Charlie* needs to **discriminate** $\{|\varphi_{x+x+}\rangle, |\varphi_{x+x-}\rangle\}$ and $\{|\varphi_{x-x+}\rangle, |\varphi_{x-x-}\rangle\}$

$$H_c |\varphi_{x+x+}\rangle = H_c \left(\frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle)_{ce} \right) = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)_{ce}$$

$$H_c |\varphi_{x+x-}\rangle = H_c \left(\frac{1}{2} (|00\rangle - |01\rangle + |10\rangle + |11\rangle)_{ce} \right) = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)_{ce}$$

$$H_c |\varphi_{x-x+}\rangle = H_c \left(\frac{1}{2} (|00\rangle + |01\rangle - |10\rangle + |11\rangle)_{ce} \right) = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)_{ce}$$

$$H_c |\varphi_{x-x-}\rangle = H_c \left(\frac{1}{2} (|00\rangle - |01\rangle - |10\rangle - |11\rangle)_{ce} \right) = \frac{1}{\sqrt{2}} (-|01\rangle + |10\rangle)_{ce}$$

- Charlie* will know that Alice measured $|x^+\rangle$ when $|00\rangle$ or $|11\rangle$ is measured, while he will know that she measured $|x^-\rangle$ when $|01\rangle$ or $|10\rangle$ is measured



Ancilla Attack: Practical Implementation (8/8)

3 HBB99: Dishonest Participant

- For all the cases Charlie* can avoid the detection and deduce Alice's secret
- Announcement-measurement relationships** for test rounds

	(i)	(ii)	(iii)	(iv)
0	10, 01	10, 11	10, 01	10, 11
1	00, 11	00, 01	00, 11	00, 01

- Secret-measurement relationships** for data rounds

	(i)	(ii)	(iii)	(iv)
0	00, 11	00, 11	10, 01	10, 01
1	10, 01	10, 01	00, 11	00, 11



Demo 4: Ancilla Attack

3 HBB99: Dishonest Participant

- **Roles:** Charlie* acts as a sophisticated attacker that uses an additive ancilla qubit
- **Configuration:**
 - **Attack Vector:** Charlie* use the ancilla to avoid detection and steal information
 - **Target:** Reconstruct Alice's key without triggering the error threshold
- **Results:**
 - The error rate in test rounds remains consistent with the baseline
 - **Outcome:** The protocol continues and Charlie* successfully computes the key of Alice, fully compromising the security



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The Fix: Commit-then-Reveal

4 HBB99: Fairness Fix

- **Root Vulnerability:** Both the *Ancilla Attack* and *Fake Entanglement Attack* rely on **delayed choice**:
 - Charlie* captures the system but **waits** for the disclosure of Alice's and Bob's MBs before acting
 - This allows him to **adapt his outcome** to the expected parity constraints for his chosen basis
- **Solution:** We enforce a strict ordering in test rounds:
 1. **Commitment:** All participants must announce their **outcome bits first**
 2. **Reveal:** Only **after** bits are committed, the **bases** are announced
- This forces Charlie* to guess the basis, turning a deterministic cheat into a probabilistic gamble



Fixing Fake Entanglement (1/2): The Scenario

4 HBB99: Fairness Fix

- **Attack Strategy:** Charlie* bridges the gap between Alice (a) and Bob (b) by forging his bit c to satisfy the global parity check equation:

$$a \oplus b' \oplus c = P_{target}$$

Solving for his bit:

$$c = a \oplus b' \oplus P_{target}$$

- **Moving Target:** The target parity P_{target} is not constant! It depends on the specific basis combination used in that round:
 - Bases $X, X, X \implies P_{target} = 0$ (Even)
 - Bases $X, Y, Y \implies P_{target} = 1$ (Odd)



Fixing Fake Entanglement (2/2): The Failure

4 HBB99: Fairness Fix

- **The Trap:** Charlie* must announce c *without* knowing the basis configuration, so he has to guess the target parity $P \in \{0, 1\}$
- **Wrong Guess Case:** If he guesses P but the true target was P' s.t. $P' \neq P$:
 - He announces $c = a \oplus b' \oplus P$
 - Alice checks: $a \oplus b' \oplus c$
 - Substitution: $a \oplus b' \oplus (a \oplus b' \oplus P) = P'$
 - Since $P \neq P'$, the check **FAILS**
- **Error Rate:** Charlie* has to guess the target parity, therefore

Error Rate = **50%**



Fixing the Ancilla Attack (1/2): The Scenario

4 HBB99: Fairness Fix

- **Constraint:** To cheat, Charlie* prepares a GHZ state $|\psi\rangle_{abce}$ involving his ancilla e , that enables him to distinguish mixed states
- **Basis Dependence:** The measurement required to distinguish these states depends entirely on Alice's and Bob's MBs, that influence how he will manipulate $|\psi\rangle_{abce}$
- **Incompatibility:** If Charlie* does the wrong operations, he will just compute the wrong outcome



Fixing the Ancilla Attack (2/2): The Failure

4 HBB99: Fairness Fix

- **The Forced Guess:** With the *Commit-then-Reveal* rule, Charlie* is forced to measure before knowing the bases, guessing a setup $B \in \{XX, XY, YX, YY\}$
- **The Collapse:** If the guessed bases B are different from the real one B' , he will make a wrong computation, ultimately being detected or deducing a wrong key of Alice
- **Error Rate:** Charlie* has to guess the used bases, therefore

Error Rate = **75%**



Demo 5: The Fairness Fix in Action

4 HBB99: Fairness Fix

- **Objective:** Prove that enforcing the "Commit-then-Reveal" order breaks the Fake Entanglement and Ancilla Attack.
- **Configuration:**
 - **Scenario:** The same code used for Demo 3 and Demo 4 are used
 - **Constraint:** In the code, Charlie* does not perform his actions looking at other parties' MBs, but choosing randomly between the possible options
- **Results:**
 - **Behaviour:** Charlie* is never able to both avoid the detectoin and deduce Alice's key
 - **Conclusion:** The attacks becomes just pure guessing, resulting in harder attacks as the length of the generate key increases



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Beyond Classical Secrets

5 Conclusions

- **Scope of Our Work: Classical Sharing**
 - We focused on HBB99 for establishing a correlated bit string (Key Distribution)
 - Here, the GHZ state is used to generate a secure key to encrypt classical messages
- **Alternative Application: Quantum Sharing**
 - The original HBB99 paper also proposed splitting *quantum* information (Teleportation-like)
 - In that scenario, Charlie can reconstruct Alice's unknown quantum state only by cooperating with Bob



Conclusions & Results

5 Conclusions

- **Vulnerability**

- We demonstrated that the standard HBB99 protocol is vulnerable to a dishonest participant
- By exploiting delayed measurements, an attacker can recover the secret without triggering the error threshold

- **Mitigation Strategy**

- Commit-then-Reveal fix enforces a strict ordering of events
- Attacker is forced to measure before knowing the MBs of the other parties

- **Security Achieved**

- Deterministic cheating is transformed into a probabilistic gamble, increasing the detection rate
- Verifiable QSS is the evolution to achieve deterministic security (based on Quantum Error Correcting Codes)



QSS: Analysis of HBB99 protocol

Thank you for listening!

References

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